ALEC XU

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TECHNICAL INTERESTS

- Efficient parameterizations and algorithms for vision and vision-language models.
- Computer vision and multimodal foundation models.
- Theoretical foundations of deep learning models.

EDUCATION

University of Michigan, Ann Arbor Aug. 2023 - Aug. 2027 (expected)
PhD Electrical and Computer Engineering
Advisor: Dr. Qing Qu
GPA: 4.00/4.00
Thesis: Theoretical Insights into Deep Learning Efficiency and Generalization

University of Michigan, Ann Arbor Jan. 2022 - Dec. 2022
MSE Electrical and Computer Engineering
Advisor: Dr. Jeffrey Fessler
GPA: 4.00/4.00

University of Michigan, Ann Arbor Sept. 2018 - Dec. 2021
BSE Electrical Engineering, Summa Cum Laude
GPA: 3.97/4.00

WORK EXPERIENCE

Apple May 2026 - Sept. 2026
Machine Learning Intern Sunnyvale, CA

- Pre-trained and fine-tuned 0.5B parameter transformer model to enable zero-shot video segmentation using sub-1M parameter U-Net for on-device applications.

MIT Lincoln Laboratory Jan. 2023 - July 2023
Co-Op Research Student Lexington, MA

- Developed novel edge detection reconstruction self-supervised learning (SSL) task for 3D axon segmentation.
- U-Nets pre-trained on proposed task achieved up to 5% improvement on downstream segmentation metrics over previous SSL tasks.

MIT Lincoln Laboratory May 2022 - Aug. 2022
Summer Research Intern Lexington, MA

- Developed novel two-stage deep learning pipeline for multi-channel signal detection and modulation recognition using YOLO object detection and DeepSig signal classification models.

PUBLICATIONS & PREPRINTS

† Equal contribution

[8] S. M. Kwon, **A. S. Xu**, C. Yaras, D. Song, L. Balzano, Q. Qu. "The Effect of Training Task Diversity for In-Context Learning through the Lens of Low-Dimensional Subspaces," **arXiv:2606.06814**, under review at **ACM Journal of Data Science**.

[7] **A. S. Xu**, C. Yaras, M. Asato, Q. Qu, L. Balzano. "Emergent Low-Rank Training Dynamics in MLPs with Smooth Activations," **NeurIPS 2026**.

[6] S. M. Kwon†, **A. S. Xu**†, C. Yaras, L. Balzano, Q. Qu. "Out-of-Distribution Generalization of In-Context Learning: A Low-Dimensional Subspace Perspective," **AISTATS 2026**. Short version in **AISTATS 2026 CauScale Workshop** (**Best Paper**).

[5] **A. S. Xu**, C. Yaras, P. Wang, Q. Qu. "Linearly Separable Features in Shallow Nonlinear Networks: Width Scales Polynomially with Intrinsic Data Dimension." **AISTATS 2026**.

[4] C. Yaras, **A. S. Xu**, P. Abillima, C. Lee, L. Balzano. "MonarchAttention: Zero-Shot Conversion to Fast, Hardware-Aware Structured Attention," **NeurIPS 2025** (**Spotlight**).

[3] **A. S. Xu**, N. I. Shamsi, L. A. Gjesteby, L. J. Brattain. "Self-Supervised Edge Detection Reconstruction for Topology-Oriented 3D Axon Segmentation and Centerline Detection," **WACV 2024**.

[2] N. I. Shamsi, **A. S. Xu**, L. A. Gjesteby, L. J. Brattain. "Improved Topological Preservation in 3D Axon Segmentation and Centerline Detection using Geometric Assessment-driven Topological Smoothing (GATS)," **WACV 2024**.

[1] **A. S. Xu**, L. Balzano, J. A. Fessler, "HeMPPCAT: Mixtures of Probabilistic Principal Component Analysers for Data with Heteroscedastic Noise," **ICASSP 2023**.

AWARDS & HONORS

J. Robert Beyster Computational Innovation Fellowship
Awarded to 3 domestic PhD students in Michigan's College of Engineering for the 2026 - 2027 academic year

Best Paper Award
AISTATS 2026 CauScale Workshop

Best Poster Runner-Up
Michigan Electrical & Computer Engineering Industry Day 2026

SKILLS

Machine learning: PyTorch, Hugging Face (transformers, diffusers, accelerate), W&B, DDP

Software tools: Python, Git, Docker, AWS, GCP, Claude Code